

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

$$I = \frac{V}{R}$$

The formula above is Ohm’s law for an electric circuit with current I, in amperes, potential difference V, in volts, and resistance R, in ohms. A circuit has a resistance of 500 ohms, and its potential difference will be generated by n six-volt batteries that produce a total potential difference of 6n volts. If the circuit is to have a current of no more than 0.25 ampere, what is the greatest number, n, of six-volt batteries that can be used?

Rationale

The correct answer is 20. For the given circuit, the resistance R is 500 ohms, and the total potential difference V generated by n batteries is 6n volts. It’s also given that the circuit is to have a current of no more than 0.25 ampere, which can be expressed as I < 0.25. Since Ohm’s law says that $I = \frac{V}{R}$, the given values for V and R can be substituted for I in this inequality, which yields $\frac{6n}{500} < 0.25$. Multiplying both sides of this inequality by 500 yields 6n < 125, and dividing both sides of this inequality by 6 yields n < 20.833. Since the number of batteries must be a whole number less than 20.833, the greatest number of batteries that can be used in this circuit is 20.